



Interest Rate Futures

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What are Interest Rate Futures?

- A financial derivative with payoff linked to future interest rates
- Highly standardized product for gaining exposure to interest rates with minimal counterparty exposure
- Traded on exchanges (e.g. CME) for different currencies (e.g. USD, EUR)
- Example: A futures contract on 3-month EUR interbank offered interest rate for the time-period 18 March 2026 – 17 June 2026
- Uses of IR Futures:
 - Speculate on the direction/ level of future interest rates
 - Hedge against movements in future interest rates
- Good liquidity is only present for 2-3 years in Europe and 4-5 years in the US

Futures on IBOR Rates

- Interbank Offered Rates (IBORs) serve as the reference rate at which banks borrow in the interbank market for different time periods (1 month, 3 months, 6 months, 12 months)
 - For example, EURIBOR for EUR borrowing
- EURIBOR Futures are quoted and traded on 3-month EURIBOR deposit rate
- IMM (International Monetary Market) dates: the 3rd Wednesday of March, June, September and December for each year
 - EURIBOR futures are on 3-month EUR deposits that start on IMM dates
 - In addition, future contracts also available for four serial months
 - For example, in April serial contracts are available for May, July, August and October delivery months in addition to quarterly contracts for June and September
- 28 delivery months in total available for trading (i.e., up to 6 years in future)

Futures on IBOR Rates

- Quote Price: $Q = (1 - R) * 100$
 - Where R is the projected 3-month EURIBOR rate
- R can be derived from the Quote Value as:
 - $R = 1 - Q/100$
 - For example, a quote value of 95.00 means the projected underlying 3-month EURIBOR rate is 5%
- Unit of trading: $€2,500 * Q$
 - Heuristically, €2,500 can be thought of as the interest amount corresponding to 1% per annum interest rate charged for 3 months on a principal of €1,000,000
- Tick size (i.e., minimum price fluctuation): 0.005
 - Tick value = $0.005 * €2,500 = €12.50$
- The daily settlement (or closing) prices are used to calculate mark-to-market of the futures contracts.

Example

- Daily Profit / Loss is computed as:
 - $P \& L = \text{€ } 2500 * \Delta Q$

	Closing Quote	Daily P & L
Day 1	94.50	0
Day 2	94.60	€ 250
Day 3	94.45	- € 375
Day 4	94.65	€ 500

Futures on USD SOFR

- Secured Overnight Funding Rate: 1-day rate underpinned by the U.S. Treasury overnight repurchase market (transactions of ~\$750 billion per day)
- SOFR Futures: 3-month future on compounded daily SOFR rate & 1-month future on averaged daily SOFR rate
- Compounded rate: $1 + R \cdot d / 360 = (1 + r_1 d_1 / 360)(1 + r_2 d_2 / 360) \dots (1 + r_n d_n / 360)$
 - Where d is the total number of days in the 3-month period
 - d_i = number of calendar days between two consecutive business days (1 if no holidays in between, 3 over a weekend, etc.)
- Averaged rate: $R = (r_1 d_1 + r_2 d_2 + \dots + r_n d_n) / d$
- Final rate determined at the end of the period, whereas for IBOR the rate is fixed at the start of the period

FRA

- Forward rate agreement (FRA) is an agreement to borrow/lend a notional cash sum for a certain period (up to 12 months) starting in future.
 - Essentially a forward starting loan except that no principal exchange is made, only the interest difference between the agreed rate and the realized rate is paid.
- Reference rate for FRA is normally the IBOR rate corresponding to the contract's currency and the length of the underlying loan.
- For a FRA referencing an IBOR rate for period (T_1, T_2) :
 - Payoff = $N \cdot DCF \cdot (R - K)$ paid at T_2
 - Where R is the IBOR rate observed at T_1 , K is the agreed rate for FRA, DCF is the “day count fraction” for the period (T_1, T_2)
 - Example: FRA contract with agreed rate $K = 5\%$ for 3-month EURIBOR for period 1 Jan 2025 – 31 March 2025 ($DCF = 90/365$), on a notional/ principal amount of €1,000,000

FRAs vs Futures

FRA	Future
Over the counter (OTC)	Exchange traded
Quoted in terms of rate	Quoted in terms of price
Settled once	Settled daily
Settled at the end of rate period	Final settlement at the start of rate period

- This difference in the settlement terms results in a gap between forward rates (implied from FRA) and the futures rates (implied from IBOR futures)
- This is referred to as convexity adjustment

Convexity Adjustment

- Futures are settled daily, and the profit/ loss is realized immediately
- When rates go down, the futures price goes up and a long position earns profit. But this profit is invested at low interest rate
- When rates go up, there is a loss which needs to be financed at higher interest rate
- To compensate for this, the future trades at a lower price. In other words, the implied interest rate from futures price is higher than the FRA par rate
- Convexity Adjustment = Future rate – Forward rate

$$\approx \frac{\sigma^2 T_1 T_2}{2}$$

- Where σ is the volatility in the interest rate (standard deviation of rate changes)
- ~1-2 basis points (bp) for 2y maturity and up to 20 bp for 5y maturity (1 bp = 0.01%).

FRA Par Rate

- Stylized example: Assume we have zero-coupon bonds traded in the market at price $P(t,T)$ yielding risk-free interest rates
- We want to find out at what rate should one enter a FRA agreement to borrow a notional amount of \$1 at T_1 and repay it at T_2 .

Replication & No-Arbitrage Arguments:

- Consider two zero-coupon risk-free bonds: Bond-1 paying \$1 at time T_1 , with current price $P(0, T_1)$, and Bond-2 paying \$1 at T_2 with price $P(0, T_2)$
- Purchase one unit of Bond-1 and sell $P(0, T_1)/P(0, T_2)$ units of Bond-2
- Net investment at current time = 0
- At T_1 when Bond-1 matures, we receive \$1. At T_2 when Bond-2 matures, we pay $P(0, T_1)/P(0, T_2)$
- Thus, using the above replication using bonds, we have borrowed \$1 at T_1 and repaid it at T_2 , with effective interest rate of $K = (P(0, T_1)/P(0, T_2) - 1)/DCF(T_1, T_2)$
- By no-arbitrage principle, an at-par FRA should be quoted at this rate

Further Readings

- Options, Futures, and Other Derivatives by John C. Hull.
- Euribor Futures details on ICE Exchange website: <https://www.ice.com/products/38527986/Three-Month-Euribor-Futures>
- SOFR Futures details on CME Exchange website: <https://www.cmegroup.com/trading/interest-rates/secured-overnight-financing-rate-futures.html>

Thank you

